The Operational Research Society of Hong Kong

The 1st ORSHK Young Researchers Workshop

jointly organised by

Operational Research Society of Hong Kong

and

Department of Industrial and Manufacturing Systems Engineering,
The University of Hong Kong

Date: 26 November 2022
Time: 09:30 to 18:00
Venue: KK Leung Building, Main Campus, The University of Hong Kong

Time	Event
09:30 - 09:45	**Registration** Venue: KK-101 (1/F)
09:45 - 10:00	**Opening Session** Venue: KK-101 (1/F) Welcome speeches by *Professor Janny M.Y. LEUNG* *President of Operational Research Society of Hong Kong (ORSHK)* *President of International Federation of Operational Research Societies (IFORS)* *College Master of Choi Kai Yau College, Professor of State Key Laboratory of Internet of Things for Smart City, and Professor of Faculty of Business Administration, The University of Macau* and *Professor Zuo-Jun Max SHEN* *Vice-President and Pro-Vice-Chancellor (Research), Dean of Graduate School, Director of HKU Musketeers Foundation Institute of Data Science, Acting Director of Technology Transfer Office, and Chair Professor in Logistics and Supply Chain Management, The University of Hong Kong*

10:00 - 12:00	**Tutorial** Venue: KK-101 (1/F) Operations Research Techniques for Data Science and Machine Learning *Professor Anthony Man-Cho SO* *Dean of Graduate School, Deputy Master of Morningside College,* *and Professor of Department of Systems Engineering and Engineering Management,* *The Chinese University of Hong Kong*	
12:00 - 13:30	Lunch Break	
13:30 - 15:45	**OR Theory I** Venue: KK-LG103 (LG/F) Session Chair: *Dr. Zhi CHEN*	**OR Applications I** Venue: KK-LG106 (LG/F) Session Chair: *Dr. Jamal Abdul NASIR*
	Provable Convergent Algorithms for Gromov-Wasserstein and Outlier Robust Gromov-Wasserstein *Lemin KONG, Jiajin LI, Jianheng TANG, Huikang LIU, Anthony Man-Cho SO*	A Multifunctional Simulation Platform for On-demand Ride Service Operations: Pricing, Matching, Repositioning *Siyuan FENG, Jintao KE, Hai YANG*
	Two-Stage Distributionally Robust Optimization with Uncertain Moments *Wen BAI, Zhi CHEN, Grani A. HANASUSANTO, Viet Anh NGUYEN*	Impact of Self-service Technology in Designing a Service Delivery System *Jie WANG, Lijun MA, Weili XUE, Yong-Hong KUO*
	Optimal Exact Recovery in the Hypergraph Stochastic Block Model via Projected Tensor Power Method *Jinxin WANG, Yuen-Man PUN, Xiaolu WANG, Peng WANG, Anthony Man-Cho SO*	Distributionally Robust Pricing with Asymmetric Information *Yihua HE, Hongqiao CHEN, Jin QI, Lianmin ZHANG*
	Robust Convex Optimization at Scale: A Max-Min-Max Framework *Kai TU, Zhi CHEN, Man-Chung YUE*	Target-Oriented Distributionally Robust Optimization and Its Applications to Surgery Allocation *Vincent Tsz Fai CHOW, Zheng CUI, Daniel Zhuoyu LONG*
	No Dimension-Free Deterministic Algorithm Computes Approximate Stationarities of Lipschitzians *Lai TIAN, Anthony Man-Cho SO*	Stability Measure for Prefab Balancing in Prefabrication Construction Supply Chain Management *Wennan ZHANG, Chenglin YU, Ray Y ZHONG*
	Nonsmooth Composite Nonconvex-Concave Minimax Optimization *Jiajin LI, Linglingzhi ZHU, Anthony Man-Cho SO*	

15:45 - 16:00	Break	
16:00 - 18:00	**OR Theory II** Venue: KK-LG103 (LG/F) Session Chair: *Dr. Wenjie HUANG*	**OR Applications II** Venue: KK-LG106 (LG/F) Session Chair: *Dr. Vincent Tsz Fai CHOW*
	Low-Pass Graph Signal Processing with Normalized Dirichlet Energy *Shangyuan LIU, Xiaolu WANG, Anthony Man-Cho SO*	Real Time Optimization of Train System Energy Consumption Via Integrated Service Scheduling And Train Speed Control *Shouyi WANG, Yanzuo LIN, Andy CHOW*
	Distributionally Robust Group Testing with Correlation Information *Daniel Zhuoyu LONG, Jin QI, Yu SUN, Aiqi ZHANG*	Equilibrium Analysis for On-Demand Food Delivery Markets *Jintao KE, Ce WANG, Xinwei LI*
	Adjustable Distributionally Robust Optimization with Infinitely Constrained Ambiguity Sets *Haolin RUAN, Zhi CHEN, Chin Pang HO*	A Home Health Care Logistics Planning Problem With Prioritized Patients And Fuzzy Demands *Jamal Abdul NASIR, Yong-Hong KUO*
	A Momentum-Accelerated Adaptive Cubic Regularization Method and Approximate Secular Equations for Cubic Regularization Subproblems *Yihang GAO, Man-Chung YUE, Michael NG*	Online Portfolio Selection with Constant Cash Inflows *Benmeng LYU, Boqian WU, Sini GUO, Jia-Wen GU, Wai-Ki CHING*
	Modularity, Convex Conjugates and MMOT *Joshua Zoen-Git HIEW*	Efficient Simulation of Polyhedral Expectations *Dohyun AHN, Lewen ZHENG*

Presenters are underlined.

Tutorial

Venue: KK-101 (1/F), Time: 10:00 - 12:00

Operations Research Techniques for Data Science and Machine Learning

Professor Anthony Man-Cho SO

Dean of Graduate School, Deputy Master of Morningside College, and
Professor of Department of Systems Engineering and Engineering Management,
The Chinese University of Hong Kong

ABSTRACT

Many techniques we learn in operations research are key to enabling various data science and machine learning applications. In this tutorial, we present several representative applications and demonstrate how tools from optimization and probability can be used to develop tractable mathematical models with performance guarantees for them.

BIO

Anthony Man-Cho So received his BSE degree in Computer Science from Princeton University with minors in Applied and Computational Mathematics, Engineering and Management Systems, and German Language and Culture. He then received his MSc degree in Computer Science and his PhD degree in Computer Science with a PhD minor in Mathematics from Stanford University. Dr. So joined The Chinese University of Hong Kong (CUHK) in 2007. He is currently Dean of the Graduate School, Deputy Master of Morningside College, and Professor in the Department of Systems Engineering and Engineering Management. His research focuses on optimization theory and its applications in various areas of science and engineering, including computational geometry, machine learning, signal processing, and statistics.

Dr. So is appointed as an Outstanding Fellow of the Faculty of Engineering at CUHK in 2019. He currently serves on the editorial boards of Journal of Global Optimization, Mathematical Programming, Optimization Methods and Software, and SIAM Journal on Optimization. He has also served as the Lead Guest Editor of IEEE Signal Processing Magazine Special Issue on Non-Convex Optimization for Signal Processing and Machine Learning. Dr. So has received a number of research and teaching awards, including the 2018 IEEE Signal Processing Society Best Paper Award, the 2015 IEEE Signal Processing Society Signal Processing Magazine Best Paper Award, the 2014 IEEE Communications Society Asia-Pacific Outstanding Paper Award, and the 2010 INFORMS Optimization Society Optimization Prize for Young Researchers, as well as the 2022 UGC Teaching Award, the 2022 CUHK University Education Award, and the 2013 CUHK Vice-Chancellor's Exemplary Teaching Award. He also co-authored with his student a paper that receives the Best Student Paper Award at the 19th IEEE International Workshop on Signal Processing Advances in Wireless Communications (SPAWC 2018).

Session: OR Theory I

Venue: KK-LG103 (LG/F), Time: 13:30 – 15:45

Session Chair: Dr. Zhi CHEN

Provable Convergent Algorithms for Gromov-Wasserstein and Outlier Robust Gromov-Wasserstein

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ABSTRACT

The Gromov-Wasserstein (GW) distance provides a flexible way to compare and couple probability distributions supported on different metric spaces. Recently, the GW distance has acted as a main modeling tool for aligning heterogeneous data for a host of graph learning tasks. We propose the first provable single-loop algorithm, called the Bregman Alternating Projected Gradient (BAPG) method, for computing the GW distance. By establishing an error bound condition, we show that BAPG asymptotically converges to a critical point of the GW problem and establish an explicit approximation bound for the distance between the fixed-point set of BAPG and the critical point set of GW.

It is known that the GW distance can be extremely sensitive to outliers when they are weighed similarly as the other samples in its objective function. To address this issue, we propose a new robust model of the GW distance (RGW) with optimistically perturbed marginal constraints. To realize its benefits in practice, we further develop a computational efficient and yet theoretically provable computational procedure based on the Bregman proximal alternating linearization minimization algorithm. We conduct extensive numerical experiments to not only corroborate our theoretical findings but also demonstrate the effectiveness of BAPG on graph matching tasks and RGW on the subgraph matching task.

Keywords: gromov-wasserstein, graph learning, nonconvex optimization

Two-Stage Distributionally Robust Optimization with Uncertain Moments

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ABSTRACT

The two-stage distributionally robust linear program under uncertain parameters is typically NP-hard and is usually solved approximately. In this paper, we consider an ambiguity set of uncertain mean and covariance and reformulate this problem as a finite-dimensional copositive program. In particular, we construct the ambiguity set by using the Wasserstein distance and the Kullback–Leibler divergence to specify the uncertain mean and covariance. To use the sample information more effectively, especially when the sample size is large, we add a weight to connect the proposed method with the classical sample average approximation approach. Through numerical studies on a lot-sizing problem, we demonstrate that our method is practically tractable and remains competitive in terms of running time and optimal solutions.

Keywords: distributionally robust optimization, uncertain moments, copositive programming, wasserstein distance

Optimal Exact Recovery in the Hypergraph Stochastic Block Model via Projected Tensor Power Method

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ABSTRACT

In this work, we investigate the problem of exact community recovery in the symmetric d-uniform ($d \geq 2$) hypergraph stochastic block model (d-HSBM), where a d-uniform hypergraph with n nodes is generated by first partitioning the n nodes into K equal-sized disjoint communities ($K \geq 2$) and then generating d-uniform hyperedges with probability that depends on the community memberships of the d nodes involved. Despite the fact that the maximum likelihood estimation problem is non-convex and discrete, we develop a simple yet efficient iterative method, called the **projected tensor power method**, to tackle it. If the initialization satisfies a partial recovery condition in the logarithmic degree regime of the problem, we show that our proposed method can recover the hidden community structure with high probability down to the information-theoretic limit. Besides, with the aforementioned initialization condition, our proposed method has a time complexity of $O(n \log^2 n / \log\log n)$, which is competitive with existing state-of-the-art methods. We also conduct numerical experiments of the developed method to validate our theoretical findings.

Keywords: nonconvex optimization, community recovery, hypergraph SBM, time complexity

Robust Convex Optimization at Scale: A Max-Min-Max Framework

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ABSTRACT

Robust optimization (RO) is a popular and powerful modeling paradigm to address real-world optimization problems that are subject to parameter uncertainty. Existing reformulation method and cutting plane method for solving RO problems do not scale well as the problem size becomes large. In this work we propose a max-min-max framework to solve an RO problem and design, by leveraging the problem's structure from this perspective, a proximal-type first-order method that returns (near) optimal solutions with an optimal iteration complexity. Numerical evidences show the encouraging performance of our approach against existing methods in terms of computational speed, when the problem size is medium and large.

Keywords: parameter uncertainty; robust optimization; max-min-max framework; first-order method

No Dimension-Free Deterministic Algorithm Computes Approximate Stationarities of Lipschitzians

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ABSTRACT

We consider the computation of an approximately stationary point for a Lipschitz and semi-algebraic function with a local oracle. If the function is smooth, simple deterministic methods have dimension-free finite oracle complexities. For the general Lipschitz setting, only recently, (Zhang et al., Proc. of ICML, 2020) introduced a randomized algorithm that computes Goldstein's approximate stationarity (Goldstein, Math Prog, 1977) to arbitrary precision with a dimension-free polynomial oracle complexity.

In this paper, we show that no deterministic algorithm can do the same. Even without the dimension-free requirement, we show that any finite time guaranteed deterministic method cannot be general zero-respecting, which rules out most of the oracle-based methods in smooth optimization and any trivial derandomization of (Zhang et al., Proc. of ICML, 2020). Our results reveal a fundamental hurdle of nonconvex nonsmooth problems in the modern large-scale setting and their infinite-dimensional extension.

Keywords: nonconvex optimization, nonsmooth optimization, lower bound, oracle complexity, derandomization

Nonsmooth Composite Nonconvex-Concave Minimax Optimization

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ABSTRACT

Nonconvex-concave minimax optimization has received intense interest in machine learning, including learning with robustness to data distribution, learning with non-decomposable loss, adversarial learning, to name a few. Nevertheless, most existing works focus on the gradient-descent-ascent (GDA) variants that can only be applied in smooth settings. In this paper, we consider a family of minimax problems whose objective function enjoys the nonsmooth composite structure in the variable of minimization and is concave in the variables of maximization. By fully exploiting the composite structure, we propose a smoothed proximal linear descent ascent (smoothed PLDA) algorithm and further establish its $O(\varepsilon^{-4})$ iteration complexity, which matches that of smoothed GDA. Moreover, under the mild assumption that the objective function satisfies the one-sided Kurdyka-Łojasiewicz condition with exponent $\theta \in [0, 1)$, we can further improve the iteration complexity to $O(\varepsilon^{-2\max\{2\theta,1\}})$. To the best of our knowledge, this is the first provably efficient algorithm for nonsmooth nonconvex-concave problems that can achieve the optimal iteration complexity $O(\varepsilon^{-2})$ if $\theta \in [0, 1/2]$. As a byproduct, we discuss different stationarity concepts and clarify their relationships quantitatively, which could be of independent interest. Empirically, we illustrate the effectiveness of the proposed smoothed PLDA in variation regularized Wasserstein distributionally robust optimization problems.

Keywords: minimax optimization, nonsmooth analysis, proximal linear algorithm

Session: OR Applications I

Venue: KK-LG106 (LG/F), Time: 13:30 – 15:45

Session Chair: Dr. Jamal Abdul NASIR

A multifunctional simulation platform for on-demand ride service operations: pricing, matching, repositioning

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ABSTRACT

Ride-sourcing services, offered by transportation network companies like Uber and Didi, have been experiencing fast development and reshaping the way people travel in the past decade. Various mathematical models and optimization algorithms, including reinforcement learning approaches, have been developed in the literature to help ride-sourcing platforms design better operational strategies to achieve higher operational efficiency. However, due to cost and reliability issues (implementing an immature algorithm for real operations may result in system turbulence), it is commonly infeasible to validate these models and train/test these optimization algorithms within real-world ride sourcing platforms. Acting as a useful test bed, a simulation platform for ride-sourcing systems will thus be very important for both researchers and industrial practitioners to conduct algorithm training/testing or model validation through trails and errors. While previous studies have established a variety of simulators for their own tasks, it lacks a fair and public platform for comparing the models/algorithms proposed by different researchers. In addition, the existing simulators still face many challenges, ranging from their closeness to real environments of ride-sourcing systems, to the completeness of different tasks they can implement. To address the challenges, we propose a novel multi-functional and open-sourced simulation platform for ride-sourcing systems, which can simulate the behaviors and movements of various agents on a real transportation network. It provides a few accessible portals for users to train and test various optimization algorithms, especially reinforcement learning algorithms, for a variety of tasks, including on-demand matching, idle vehicle repositioning, and dynamic pricing. In addition, it can be used to test how well the theoretical models, developed in the literature for equilibrium analysis and strategic planning, approximate the simulated outcomes. Evaluated on real-world data based experiments, the simulator is demonstrated to be an efficient and effective test bed for various tasks related to on-demand ride service operations.

Keywords: traffic simulation, on-demand service, machine learning, urban computing

Impact of Self-service Technology in Designing a Service Delivery System

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ABSTRACT

Self-service technologies (SSTs) have been widely adopted in industries that require the delivery of physical products by services, in which consumers evaluate a product for both the product value and the service value. A typical service delivery system usually involves sales agents and/or self-service technologies, for example, online services and kiosks, to serve consumers by a coproduction process. In other words, both the sales agent (or self-service machine) and the consumer should exert effort, with corresponding service costs. By modeling the coproduction output with a Cobb–Douglas production function, we establish a principal–agent model to study the value of self-service technologies in designing a service delivery system wherein the sales agent's service cost is private information. We first characterize the main trade-off between the sales agent and the self-service machine when the firm provides only one service channel. Then, we analyze the value of the self-service machine when the firm can provide both service channels. We find that, interestingly, the firm may possibly provide both service channels, that is, services offered by the sales agent and the self-service machine, when the level of information uncertainty is high, and the self-service machine's service cost is intermediate. Moreover, when both service channels are offered, only the efficient sales agent will be contracted, and the inefficient sales agents are screened out of the market by the self-service machine; that is, the self-service machine can help the firm eliminate the information rent. We also investigate how the firm's service weight in the coproduction process and information uncertainty influence the consumer surplus, firm's choices, contract parameters, and resulting profits. Our results are shown to be robust when our model is extended to consider a single-contract strategy, contracting on effort, a continuous sales agent type, and a general coproduction function.

Keywords: contract design, coproduction system, self-service machine, service delivery

Distributionally Robust Pricing with Asymmetric Information

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ABSTRACT

Traditional monopoly pricing models assume full knowledge of the expected demand for target customers. In practice, however, this over-reliance on strict assumptions and wrong estimation of model parameters may lead to significant revenue loss and misleading selling policies. In this paper, we study a distributionally robust pricing model based on asymmetric information of the customers' valuation distribution by utilizing semivariance information. We utilize the strong duality theorem to derive the tight lower bound and upper bound of $P(V \geq p)$ in a function of mean, variance, and semivariance similar to one-sided Chebyshev's inequality. Moreover, we show a series of asymptotic three-point distributions gradually approaching the probability. Then we provide the tight lower bound of the seller's profit subsequently. By maximizing the profit lower bound, we obtain the optimal robust price in a closed form. With a provable higher performance guarantee, our model outperforms the robust model with known mean and variance in all possible scenarios. Moreover, by numerical experiments, we show that our pricing strategy can obtain more than 95% maximal revenue for information goods under some common underlying distributions, which leads to huge practical value. Finally, we construct the symmetric robust pricing model under symmetric distributions and elicit its advantages against the original models.

Keywords: pricing, semivariance, distributionally robust optimization, performance

Target-Oriented Distributionally Robust Optimization and Its Applications to Surgery Allocation

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ABSTRACT

In this work, we propose a decision criterion that characterizes an enveloping bound on monetary risk measures and is computationally friendly. We start by extending the classical value at risk (VaR) measure. Whereas VaR evaluates the threshold loss value such that the loss from the risk position exceeding that threshold is at a given probability level, it fails to indicate a performance guarantee at other probability levels. We define the probabilistic enveloping measure (PEM) to establish the bound information for the tail probability of the loss at all levels. Using a set of normative properties, we then generalize the PEM to the risk enveloping measure (REM) such that the bound on the general monetary risk measures at all levels of risk aversion are captured. The coherent version of the REM (CREM) is also investigated. We demonstrate its applicability by showing how the coherent REM can be incorporated in distributionally robust optimization. Specifically, we apply the CREM criterion in surgery block allocation problems and provide a formulation that can be efficiently solved. Based on this application, we report favorable computational results from optimizing over the CREM criterion.

This project studies a fundamental problem in operations research: what criteria to optimize when uncertainties are involved. Extending from the classical chance constraint model, we propose a new decision criterion by an axiomatization approach. We then investigate the computing issue in the corresponding distributionally robust optimization problem. In particular, we provide solution methods for continuous and discrete optimization. After that, we apply it to a practical operations problem, surgery allocation decisions in healthcare management. The computational studies demonstrate the appealing performance of our proposed approach on this surgery allocation problem.

(This work is published on *INFORMS Journal on Computing*.)

Keywords: risk measure, enveloping bound, distributionally robust optimization, surgery block allocation

Stability Measure for Prefab Balancing in Prefabrication Construction Supply Chain Management

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ABSTRACT

Prefabrication construction supply chain management measures comprise tactical operating procedures to minimize costs and strategically allocate the resources such as prefab and construction materials. However, current practices in prefab allocation depend mainly on foreman experience, resulting in unpunctual delivery and insufficient resource allocation. As a costly component, the storage and distribution of prefabs have received considerable attention. The new challenges derive from coordinating the punctual transportation of prefabs and cost control. Existing literature rarely investigates resource allocation in prefabrication construction supply chain management. This research presents a prefab resource allocation model with a penalty mechanism to balance the stability of prefabrication construction supply chain management. Several aspects of practical relevance are considered, such as the penalty of unpunctual delivery, limitation of storage area in the construction site, and multiple prefab types and transportation routes. In this research, CPLEX is used for solving this mixed integer linear programming model on a small scale. A PSO based on CPLEX is proposed to solve large-scale instances with a penalty mechanism. The results show that the conduct of the penalty mechanism could save the cost of the main contractor.

Keywords: prefabrication construction supply chain management, resource allocation, penalty mechanism, particle swarm optimization

Session: OR Theory II

Venue: KK-LG103 (LG/F), Time: 16:00 – 18:00

Session Chair: Dr. Wenjie HUANG

Low-Pass Graph Signal Processing with Normalized Dirichlet Energy

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ABSTRACT

Graph signals are data defined on irregular graph domains and are ubiquitous in networked systems. It is traditionally assumed that graph signals are *smooth* over the networks on which they are defined. However, the recent graph signal processing (GSP) literature suggests that many real-world graph signals can be viewed as the outputs of *low-pass graph filters*. In spite of the popularity of this perspective, there still lacks a quantitative measure of how low-pass a graph signal is, which hinders our understanding of a host of GSP tasks. In this paper, we propose a principled yet simple metric, called the *normalized Dirichlet energy (NDE)*, to measure the low-passness of a graph signal. With this metric, we theoretically demonstrate that signal smoothness and low-passness are essentially different. Moreover, we show by spectral analysis that the NDE serves as an interpretable metric that reflects the effect of low-pass filtering. The distinctions between low-pass graph signals and other graph signals necessitate the development of customized models for various low-pass GSP tasks. In particular, we formulate graph learning and graph signal denoising problems using our NDE metric and develop efficient algorithms to solve them. Numerical experiments on both synthetic and real data demonstrate that the NDE-based models enable us to better extract information from low-pass graph signals.

Keywords: low-pass graph signal processing, graph learning, graph signal denoising, spectral analysis

Distributionally Robust Group Testing with Correlation Information

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ABSTRACT

Group testing is an efficient method to conduct massive tests using limited resources. We consider a distributionally robust group testing problem to derive an optimal partition policy by minimizing the worst-case systematic objective function, which is the weighted summation of the expected false-negative classifications, the false-positive classifications, and the number of total tests. Different from the literature, our model considers the correlation among the infection status of subjects in a population. Specifically, our ambiguity set considers prevalence and correlation information. Assuming subjects are exchangeable in the population and considering pairwise correlation, we solve the inner problem and give the closed-form of the worst-case distribution. Leveraging the information of the worst-case distribution, we prove the structural property of the optimal partition policy, that all groups are of the same size, and develop an efficient algorithm to identify the optimal common group size. Considering partial correlation, we focus on a two-cluster special case and compare two representative partitions to give managerial insights. We find that it will intend to test subjects from the same cluster together under a small prevalence rate and large correlation coefficient. Numerically, we compare our model with the widely-used independent model in the literature. The result shows that our model will increase the group size and save resources.

Keywords: distributionally robust optimization, group testing, healthcare, operations management

Adjustable Distributionally Robust Optimization with Infinitely Constrained Ambiguity Sets

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ABSTRACT

We study adjustable distributionally robust optimization problems where their ambiguity sets can potentially encompass an infinite number of expectation constraints. Although such ambiguity set have great modeling flexibility in characterizing uncertain probability distributions, the corresponding adjustable problems remain computationally intractable and challenging. To overcome this issue, we propose a greedy improvement procedure that consists of solving, via the (extended) linear decision rule approximation, a sequence of tractable subproblems—each of which considers a relaxed and finitely constrained ambiguity set that can be iteratively tightened to the infinitely constrained one. Through three numerical studies of adjustable distributionally robust optimization models, we show that our approach can yield improved solutions in a systematic way for both two-stage and multi-stage problems.

Keywords: adjustable optimization, distributionally robust optimization, infinitely constrained ambiguity set, linear decision rule

A Momentum-Accelerated Adaptive Cubic Regularization Method and Approximate Secular Equations for Cubic Regularization Subproblems

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ABSTRACT

The cubic regularization method (CR) and its adaptive version (ARC) are popular Newton-type methods in solving unconstrained non-convex optimization problems, due to its global convergence to local minima under mild conditions. The main aim of the work is to develop a momentum accelerated adaptive cubic regularization method (ARCm) to improve the performance. With the proper choice of momentum step size, we show the global convergence of ARCm and the local convergence under the KL property. Such global and local convergence can also be established when inexact solvers with low computational costs are employed in the iteration procedure. At each iteration, CR and its variants solve a cubic regularization subproblem (CRS). One way to solve the CRS relies on solving the secular equation, whose computational bottleneck lies in the computation of all eigenvalues of the Hessian matrix. Here we propose and analyze a novel CRS solver based on an approximate secular equation, which requires only some of the Hessian eigenvalues and is therefore much more efficient. Two approximate secular equations (ASEs) are developed. For both ASEs, we first study the existence and uniqueness of their roots and then establish an upper bound on the gap between the root and that of the standard secular equation. Such an upper bound can in turn be used to bound the distance from the approximate CRS solution based ASEs to the true CRS solution, thus offering a theoretical guarantee for our CRS solver.

Keywords: adaptive cubic regularization, momentum, KL property, inexact solvers, approximate secular equations

Modularity, Convex Conjugates and MMOT

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ABSTRACT

We developed a method that provides an option implied model independent price bounds for some multi-period multi-asset derivatives based on multi-marginal martingale optimal transport problem. We show that the distribution that set the bounds of the no-arbitrage price of the caps of a sub-modular payoff is uniquely determined. The optimal joint-distribution of those two assets at period one is the unique monotone coupling of the corresponding marginals. We interpret this as follow: in the world that the price attends the optimal bounding value(s), there is a single underlying factor that governing those two assets in the first period. Applications include robust pricing of multi-period basket options, digital option, call option on the minimum among multiple assets and valuation of multi-period illiquid currency cross rate derivatives. Since our result is based on the dual side of the multi-marginal martingale optimal transport, we also provide a hedging strategy to realize the optimal price bounds. We propose a numerical method that based on neural networks with penalization for calculating the optimal distribution and the corresponding trading strategy. We have performed some numerical examples to demonstrate our structural results with some simple marginals. This is joint work with Tongseok Lim, Brendan Pass and Marcelo Cruz de Souza.

Keywords: robust pricing and hedging, multi-period multi-asset derivatives, multi-marginal martingale optimal transport, submodular function, neural networks

Session: OR Applications II

Venue: KK-LG106 (LG/F), Time: 16:00 – 18:00

Session Chair: Dr. Vincent Tsz Fai CHOW

Real Time Optimization of Train System Energy Consumption Via Integrated Service Scheduling And Train Speed Control

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ABSTRACT

This paper aims to develop a real time optimization framework for minimizing train system energy consumption with consideration of service capacity and regularity via service scheduling and train speed control. The system consists of a centralized algorithm for optimizing train service schedules and distributed algorithms for optimizing speed profiles of individual trains. Furthermore, it is able to adaptively revise the target arrive time of the current train to the next station based on the connection between the current train and the preceding train. When a delay occurs, the system can recalculate and implement the optimal service schedule and speed profile for the current train in real time according to the revised arrive time, thereby avoiding train congestion or even collisions, while saving energy consumption. With an objective to reduce computing time for real time applications, the solution algorithm will be developed with use of surrogate-assisted cross entropy method. A virtual testbed of East Rail Line (EAL) in MTR will be developed on the refined OpenTrack micro-simulation system for testing the proposed optimization algorithms. The optimization framework will also be benchmarked against a range of established meta-heuristic methods. The study facilitates the system operator to deliver effective, reliable, and eco-efficient urban train services with emerging information and optimization technology.

Keywords: metro operations, surrogate function, eco-efficiency

Equilibrium Analysis for On-Demand Food Delivery Markets

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ABSTRACT

With the popularity of smartphones and the mobile internet, on-demand food and grocery delivery services have experienced rapid development in recent years. A food delivery platform dispatches crow-sourced drivers to pick up meals at merchants, e.g, restaurants, grocery supermarkets, and then deliver the meals to customers. In each round of dispatching, the platform may assign a driver to pickup a batch of meals collectively and then deliver them to customers sequentially. Apart from setting the delivery price paid by customers and the wage offered to drivers, the platform can also adjust the number of customers' orders assigned to one driver in each batch (denoted as assignment factor) to affect the market equilibrium and achieve certain objectives. This paper develops a mathematical model to characterize the equilibrium state of an on-demand food delivery market, and investigate the platform's optimal operating strategies in terms of the delivery price, wage and assignment factor for maximizing the platform profit or social welfare. In particular, we consider two market scenarios depending on whether the meals are ready before the drivers arrive at the merchants' locations for picking up the meals. It is interesting to find that the platform will choose a strategy leading to the scenario where the meals are ready as the drivers' arrivals at the merchants, such that the customers' expected waiting time is irrelevant of the meal preparation time. This contrasts with some previous studies assuming that the customers' waiting time is subject to the meal preparation time at the optima. Following this observation, we further discuss the main properties of the monopoly optimum, social optimum and Pareto-efficient solutions in the on-demand food delivery market.

Keywords: food delivery service, on-demand, equilibrium, revenue management, pareto-efficient frontier

A Home Health Care Logistics Planning Problem With Prioritized Patients And Fuzzy Demands

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ABSTRACT

In this paper, we study a real-world home health care (HHC) supply chain network problem with fuzzy demands and prioritized patients. Due to the growing HHC demand and the occurrence of natural adversities or pandemics (e.g., COVID-19), HHC providers frequently experience disruptions in the HHC supply chain network and a short supply of caregivers to meet the high demand. Therefore, consideration of real-life characteristics such as patient priorities, uncertain demand quantities and a readily adoptable HHC structure in response to natural adversities is essential to maintain the smooth operations of such services. To address such challenges, we propose a multi-depot (mobile health facilities (MHFs)), and multi-period HHC location-allocation and routing problem considering the medical supplies (medications, vaccines, etc.) quantities as fuzzy variables. The problem is modeled as a fuzzy chance-constrained programming model that schedules the patients considering patient priorities based potential healthcare costs, time windows, precedence and capacity constraints. A three-phased solution method, including stochastic simulation, is used to solve the problem. First, a good initial solution is developed by exploiting the problem-specific characteristics to cluster patients and allocating the clusters to appropriate MHFs, which is then improved through a hybrid simulated annealing algorithm. We conducted experiments using a realistic case in the context of Hong Kong to test the proposed HHC structure and solution method, as well as to examine the model's sensitivity to two important parameters that significantly impact total cost, the dispatcher preference index (DPI) and the assignment preference index (API). The experimental results show that the proposed model is effective in determining critical decisions for the studied HHC structure, and the proposed solution method efficiently delivers exceptional solutions.

Keywords: home healthcare, location-routing problem, prioritized patients, time windows, fuzzy demand

Online Portfolio Selection with Constant Cash Inflows

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ABSTRACT

Portfolio selection is a fundamental issue in finance that seeks the best wealth allocation strategy among a set of assets. As an application of machine learning techniques in portfolio selection, online portfolio selection (OLPS) has received increasing attention in recent years. There are numerous efficient algorithms that have been studied both theoretically and empirically. The limitation of existing online portfolio selection algorithms considering transaction costs is that the transaction remainder factor is usually derived approximately rather than precisely. This may induce an inaccurate transaction cost and suboptimal investment strategy. To tackle this problem, we introduce a novel method to deal with transaction costs and solve the precise transaction remainder factor and optimal investment strategy simultaneously for each period. We also consider an open-end fund, which allows for constant cash inflows, and formulate a general framework for online portfolio selection. Based on this novel model, we design a new OLPS algorithm with transaction costs and constant cash inflows (TCI) to maximize the cumulative wealth. Numerical experiments show that this algorithm is able to handle transaction costs and constant cash inflows effectively and also outperforms several state-of-the-art online portfolio selection algorithms.

Keywords: online portfolio selection, cash inflow, linear programming, transaction cost, robust optimization

Efficient Simulation of Polyhedral Expectations

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ABSTRACT

We consider the problem of estimating the expectation over a convex polyhedron specified by a set of linear inequalities. This problem encompasses a multitude of financial applications including systemic risk quantification, portfolio management, and option pricing. We particularly focus on the case where the target event is rare which corresponds to, for example, extreme systemic failures, large portfolio losses, and deep out-of-the-money options in the aforementioned applications, respectively. This rare event setting renders the naive Monte Carlo method inefficient and requires variance reduction techniques. Assuming that random vectors follow normal variance mixture distributions, which include multivariate normal distribution and multivariate t distribution, we develop a novel conditional importance sampling (CIS) method for the computation of the said expectation by exploiting the geometry and concentrating the sampling density almost exclusively inside the polyhedral set. We prove the bounded relative error of the CIS estimator under a general rare event regime which embraces a wide range of asymptotic regimes considered in the literature. The proposed method significantly outperforms the existing approaches in various numerical experiments in terms of accuracy and computational burden.

Keywords: rare-event simulation; importance sampling; polyhedral theory; risk management

Campus Map

Path from HKU station Exit A2 to the workshop venue

Catering outlets around the workshop venue:

- Chong Yuet Ming Amenities Centre Restaurant, 4/F, Chong Yuet Ming Cultural Centre
- Chong Yuet Ming Amenities Centre Cafeteria - Café 330, 2/F, Chong Yuet Ming Amenities Centre
- Fong Shu Chuen Amenities Centre Restaurant, 2/F, Fong Shu Chuen Amenities Centre
- Halal Food Corner - 27 Kebab House, 1/F, Fong Shu Chuen Amenities Centre
- Outpost II – SUBWAY, Run Run Shaw Podium
- Catering Outpost III - Kiosk by The Sandwich Club, Run Run Shaw Podium
- Main Library (Old Wing) Coffee Shop - The Coffee Academics, G/F, Library Building (Old Wing)

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